Background-Sector Phase-Space Sampling for the DCS-Monitor

64-Cell Pilot and HPC Validation Plan

DCS-Monitor simulation notes

October 30, 2026

Abstract

The DCS-Monitor simulation must estimate the detector response to radiation originating in neighbouring spent-fuel casks, not only in the target cask. A direct full-array transport calculation is unnecessarily expensive because the same cask-to-sector phase space is reused for many detector and shield configurations. This report describes a two-stage factorised workflow: a pilot-scale `p1.5` production step scores the first entry of each gamma or neutron into a 64-cell shell around the target cask, and a `p2_bkgd` replay step transports those unit-weight entries through the detector geometry. The shell is divided into eight azimuthal sectors and eight longitudinal layers spanning the cask height from −2040 mm to 2040 mm. It lives in a Geant4 parallel world, so it has no material and does not alter the mass geometry. The replay deduplicates entries by source event: for each source event it keeps the entry with the earliest global time, then the smaller cell identifier. This removes repeated crossings of the artificial cell boundaries, but it also retains only one candidate entry per event, so it is an approximation rather than a track-level transport of every crossing. A 662 keV gamma check against a direct ZLN calculation for cask 4 and global fuel 0 gives 961 detector hits and 208.740 MeV in the 64-cell two-cell replay, compared with 1003 hits and 209.756 MeV from the direct ZLN result scaled from 180 s to 36 s. The test contains about 953 independent hit events, so the ratios of 0.958 in hit count and 0.995 in deposited energy are consistent with the direct result at the approximately 3% statistical level. A four-cell variant gives 1016 hits and 212.392 MeV, bracketing the direct yield. The report also sets out the production matrix, merging requirements, and the HPC validation plan needed for a physics result.

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1 Motivation and scope

The nominal DCS-Monitor simulation starts from the target cask and follows radiation through the cask body, the detector shield, and the detector. That approach is appropriate for the target cask, but the detector also sees radiation produced in the other eleven casks in the storage hall. Repeating the full source-to-detector transport for every neighbouring cask, hexant, detector position, and shield variant would multiply the cost by the number of configurations.

The workflow in this report factorises the problem at a fixed geometric surface. The expensive cask-to-surface transport is performed once per source configuration, and the resulting phase space is reused for the cheaper surface-to-detector transport. The factorisation is exact only in the sense of phase-space sampling: it assumes that a recorded entry is upstream of the detector for the relevant particles and that the shell encloses the detector and shield volume. The event-level deduplication described in Section 3.3 is a second approximation. Both assumptions are part of the validation plan.

This report is deliberately a pilot report. The numerical results use one gamma line, one source cask, and a one-million-event subset of the source file. They demonstrate that the implementation and the bookkeeping are consistent; they are not a final background estimate.

2 Geometry and source model

2.1 Storage hall and detector positions

The hall contains twelve CASTOR 440/84 casks arranged as three clusters: a central 2×2 group and two four-cask side groups. The target cask is cask 0 at $(x, y) = (1580\,\text{mm}, 1580\,\text{mm})$. The concrete floor and the two side walls are included in the mass geometry. The detector scan uses eight positions at 45° intervals, nominal radius 1350 mm, and $z = 150\,\text{mm}$. The hemi-shield variant uses a larger effective radius of about 1445 mm. Figure 1 shows the plan view and the relation between the hall, the cask array, the detector positions, and the scoring shell.

2.2 The 64-cell scoring shell

The shell is divided into eight non-overlapping 45° azimuthal sectors and eight longitudinal layers. Its inner radius is 1325 mm, just outside the fin tips of the CASTOR body; its outer radius is 1775 mm; and its longitudinal extent is −2040 mm to 2040 mm, in the target-cask frame. The eight layers therefore have a thickness of 510 mm. The parameters are collected in Table 1. The shell lives in the `SectorScoringWorld` parallel world and therefore has no material. It changes neither the transport nor the detector response.

The cell identifier is

$$\texttt{cellId} = \texttt{zLayer} \times 8 + \texttt{sector}, \tag{1}$$

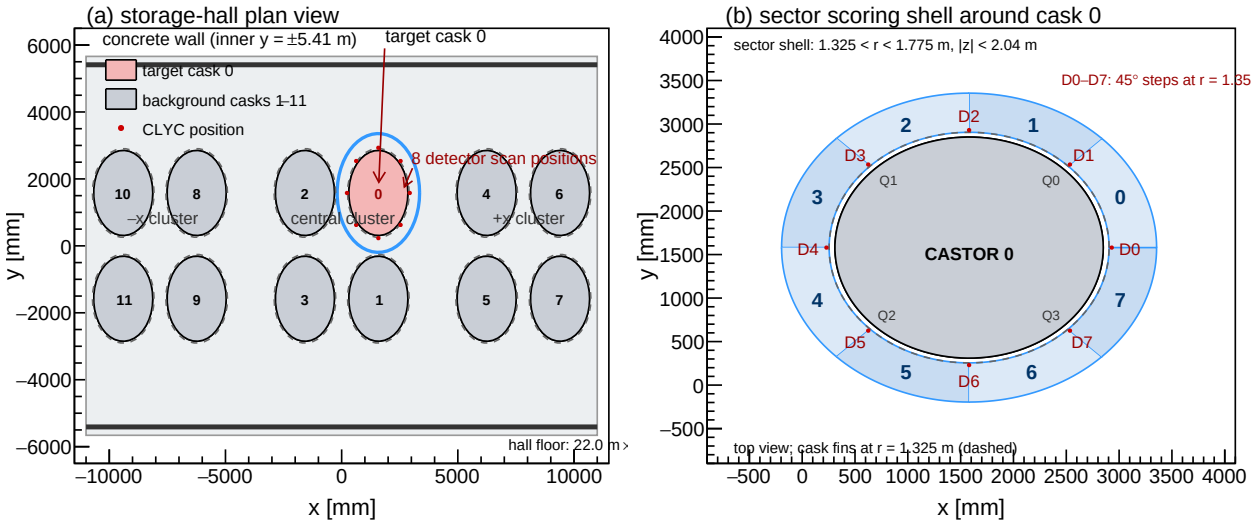


Figure 1: Storage-hall plan view (left) and the 64-cell scoring shell around target cask 0 (right). The shell is drawn in the cask-local frame; it is a scoring surface, not a material volume. The detector positions D0–D7 are shown at their nominal 1350 mm radius inside the shell.

Table 1: Reference geometry used by the 64-cell sector scoring and the 662 keV gamma validation.

Quantity	Value	Note
Casks in hall	12	three clusters of four
Target cask position	(1580 mm, 1580 mm)	cask 0
Cask body outer radius	1270 mm	fin-tip radius $\approx$ 1325 mm
Sector inner radius	1325 mm	at the fin tips
Sector outer radius	1775 mm	encloses the detector at 1350 mm
Sector z extent	-2040 mm to 2040 mm	cask height
Azimuthal sectors	8	45° each, non-overlapping
Longitudinal layers	8	510 mm each
Scoring cells	64	`zLayer` $\times$ 8 + `sector`
Entry faces per cell	6	inner, outer, two caps, two sides
Detector scan radius	1350 mm	nominal, $z = 150$ mm
Hemi-shield effective radius	$\approx$ 1445 mm	for the shield variant

so the first eight identifiers are the eight azimuthal sectors of the bottom layer and the last eight are those of the top layer. A detector at $z = 150$ mm lies in layer 4, which covers $z = 0$ mm to 510 mm. The nominal detector at $\varphi = 0°$ lies on the boundary between sectors 0 and 7, so the automatic cell selection described in Section 3.3 returns both sector cells in layer 4: cell 32 and cell 39.

Each first entry is assigned to one of the six boundary faces by the smallest distance to the inner cylinder, the outer cylinder, the top cap, the bottom cap, or one of the two azimuthal side planes. Faces 0–3 are the cylindrical and cap surfaces; faces 4 and 5 are the lower- and upper-φ side faces of the cell. For cells that are not at the bottom or top of the shell, the two longitudinal faces are internal boundaries shared with the adjacent layer; they are labelled 6 and 7 so that they can be separated from the two external caps. The face label is not used in the transport. It is used to verify that the shell is populated through all expected surfaces and to diagnose boundary duplicates.

Figure 2 shows the radial–longitudinal cross-section. The shell encloses the detector: the inner face is just outside the cask at 1325 mm, the detector sits at the nominal 1350 mm scan radius, and the outer face is at 1775 mm. Inner-face entries are therefore upstream of the detector, while outer, top, and bottom entries are particles entering from the surrounding hall. The two azimuthal side faces carry particles crossing between neighbouring sectors; they are retained as local boundary sources for the replay. The shell now spans the full cask height, so the cask top and bottom are

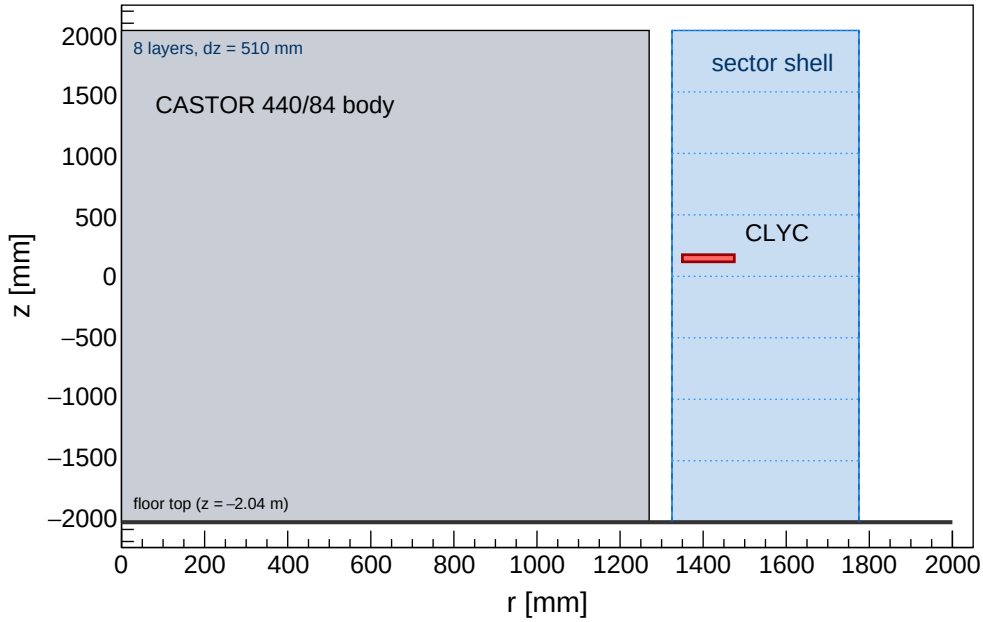


Figure 2: Radial–longitudinal view of the target cask, the 64-cell scoring shell (1325 mm $< r <$ 1775 mm, -2040 mm $< z <$ 2040 mm), and the detector at $z = 150$ mm. The horizontal dashed lines mark the eight 510 mm layers. The shell is a parallel-world scoring volume.

not truncated by the scoring surface. The eight longitudinal layers localise an entry to a 510 mm interval and prevent a single cell from covering both the detector and the cask ends.

2.3 Visual check of the array

The OpenGL views in Figure 3 are a quick geometric check of the twelve-cask array and the target detector position. The hall is omitted in these renders so that the cask bodies are not obscured by the semi-transparent walls; the hall outline is shown in Figure 1. Internal cask details are hidden in the renders to keep the exported vector files manageable. These images are illustrative, not a substitute for the dimensioned plan view.

2.4 From p1 surface flux to p1.5 sector flux

The existing p1 stage records particles crossing the outer surface of a cask. Each entry contains the particle type, kinetic energy, position, direction, and the statistical weight from the original source calculation. The p1.5 stage samples those crossings, applies the same surface smearing as the nominal p2 workflow, and transports the resulting particles in a geometry that contains all twelve casks but no detector. The 64-cell shell records the first entry of each gamma or neutron into each cell.

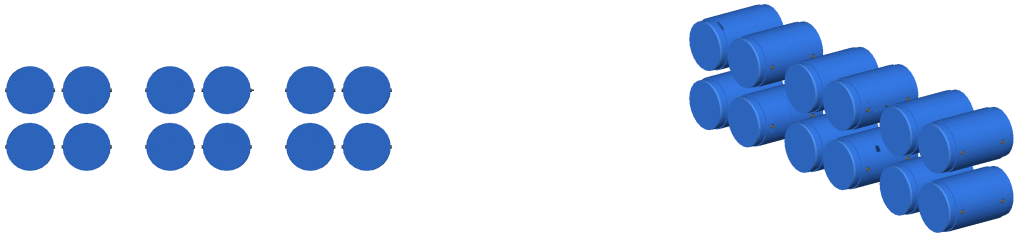
For a p1 file with N_{p1} generated primaries and total kept weight W_{kept}, the weight ratio is

$$r = \frac{W_{\mathrm{kept}}}{N_{p1}}. \tag{2}$$

For a requested generation time T_{gen} and a source decay rate λ, the p1.5 driver computes

$$N_{p1.5} = \lambda\, T_{\mathrm{gen}}\, r, \tag{3}$$

clamped to the `/run/beamOn` ceiling when necessary. The value of `numPrimaries` sets the number of source events read from the p1 file; it does not set the number of events that reach the shell. The output `sectorFlux` tree has M entries for a run. Each entry is stored with unit weight because the p1 weights have already been folded into the sampling. The provenance is written once to the `sectorMeta` tree: T_{gen}, M, $N_{p1.5}$, r, the source cask, the hexant, the source type, the target cask,



(a) Top view of the twelve casks.

(b) Oblique view.

Figure 3: Geant4/OpenGL renders of the cask array with the target detector included. The hall is hidden for clarity. The top view shows the six cask columns (two rows); the oblique view shows the three clusters in perspective.

the shell dimensions, the number of azimuthal sectors, the number of longitudinal layers, and the number of cells.

The important bookkeeping rule is that M is the total number of `sectorFlux` entries in the file, not a per-cell count. A single p1.5 run therefore writes one metadata row whose M is the sum over the cells that were populated. A merged production file must either carry one row per cell or add an explicit per-cell count; the replay sampler computes the filtered count M_{loaded} and the deduplicated count M_{unique} for the requested cell selection.

3 Implementation

3.1 Parallel world and first-entry scoring

The scoring geometry is a `G4VUserParallelWorld`. The mass geometry is constructed first; the parallel world then places $8 \times 8 = 64$ `G4Tubs` cells at the target-cask position and orientation. Because the cells have no material, they are invisible to the physics processes. The sensitive detector is attached only to the parallel-world logical volumes.

A naive implementation would record every step inside a cell, which would count a track many times and would make the entry multiplicity depend on the step limiter. The implementation instead records the first entry of each track into each cell. A thread-local registry maps the track identifier to a 256-bit mask. When a track is created, it inherits its parent's mask; when it crosses a cell boundary for the first time, the corresponding bit is set and the entry is recorded. The track is not killed. This preserves the full shower and allows a secondary produced in one cell to be counted when it later enters a different cell.

Only gammas (PDG 22) and neutrons (PDG 2112) are recorded, matching the p1 surface-flux source. The step must have `fGeomBoundary` status, so a step that merely starts inside a cell because of a geometry boundary is not mistaken for a fresh entry. The recorded position is the pre-step point, and the direction is the unit momentum at that point. The cell identifier is the copy number of the parallel-world physical volume; the sector and longitudinal layer are decoded with Equation (1).

The scoring shell is registered only when the executable is started in background-sector mode (`--mode background-sector`). This is not a convenience flag: the parallel world must be registered before the macro executes, because the run manager builds the parallel geometry during initialisation. The mode is analog, so combining it with `--biasing` is rejected.

3.2 p1.5 production

A p1.5 run uses the existing surface generator in `CASTOR440_surface_from_TTree` mode. The p1 source file is loaded once, the requested cask and hexant are selected, the surface smearing is applied, and the particles are transported through the full cask array. The detector and its shield are absent. The output contains the `sectorFlux` tree and the `sectorMeta` provenance row.

The validation macro sets `/dcs-monitor/det/setSectorZLayers 8` before `/run/initialize`, loads the full one-million-event gamma subset, and lets the decay rate and generation time determine the primary count. The target cask is present but passive; the source cask is selected with `/surf/caskNum`, and the hexant is selected with `/surf/sourceRotZ`. The source type, global fuel index, and batch identifier are copied into every entry and into the metadata.

3.3 p2_bkgd replay and event deduplication

The replay source is a separate generator mode, `BACKGROUND_SECTOR_from_TTree`. It loads the `sectorFlux` tree, optionally filters on cell identifier or longitudinal layer, and samples entries uniformly. The sampler stores only the quantities needed by the generator: PID, kinetic energy, time, position, and unit direction. The positions are already in the global frame, so no cask translation or rotation is applied at replay time.

The cell filter can be given explicitly with `/dcs-monitor/bkg/cellIds`, or it can be derived from the detector geometry with `/dcs-monitor/bkg/autoCells true`. The automatic selection builds the oriented detector envelope, transforms its corners into the target-cask frame, and selects every cell whose radial, azimuthal, and longitudinal intervals overlap the envelope. For the nominal CLYC assembly it selects cells 32 and 39, the sector-0 and sector-7 cells of layer 4. The inclusive layer overlap is deliberate: a detector that touches a layer boundary must not lose entries from the adjacent layer.

The replay applies a per-event deduplication before sampling. The key is

$$(\texttt{sourceKey}, \texttt{evtNb}), \tag{4}$$

where `sourceKey` packs the source cask, source type, hexant, global fuel, and batch identifier, and `evtNb` is the p1.5 event number. For each key the sampler keeps the entry with the smallest global time; if two entries have the same time, it keeps the one with the smaller cell identifier. The counts are reported as

$$M_{\text{loaded}} \geq M_{\text{unique}} = M, \tag{5}$$

where M_{loaded} is the number of entries after the cell filter and before deduplication. For a replay of duration T_{replay}, the number of generated primaries is

$$N_{\text{replay}} = M_{\text{unique}} \frac{T_{\text{replay}}}{T_{\text{gen}}}. \tag{6}$$

The replay sets the particle weight to one; it must not apply the p1 or p1.5 weight again.

The deduplication is not track-aware. A source event that enters two selected cells contributes one entry, not two, so the replay is a sample of one candidate entry per event rather than a sum over all cell crossings. This removes the double counting that occurs when the same source particle crosses the sector-0/7 boundary, but it can discard a later entry that is geometrically closer to the detector. The choice of the earliest entry is a modelling approximation. Section 6 states the checks needed before it is used for a production background estimate.

3.4 Workflow

Figure 4 summarises the data flow. The upper branch is the background workflow; the lower branch is the existing target-cask workflow, which remains unchanged. The two branches meet only at the detector-hit and analysis level.

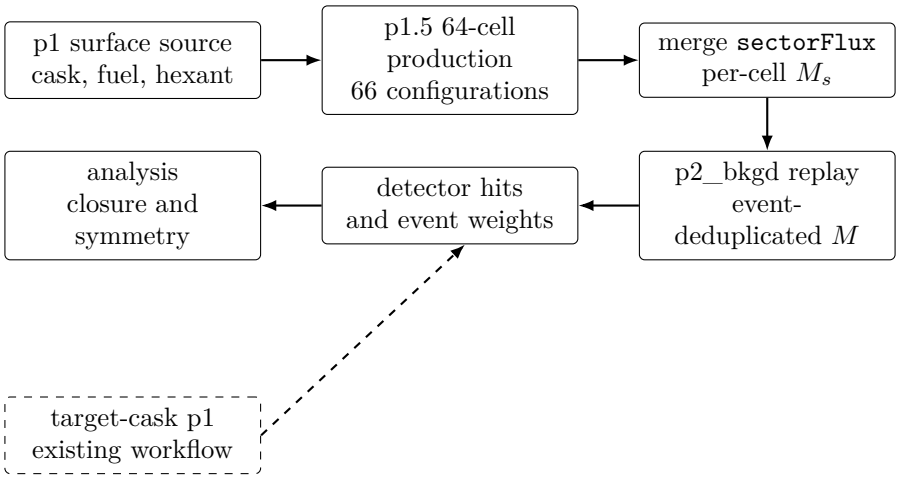


Figure 4: Two-stage background workflow. The p1.5 stage performs the expensive cask-to-cell transport once per source configuration; the p2_bkgd stage reuses the unit-weight phase space for detector and shield variants. The dashed path is the existing target-cask workflow.

4 Fast 662 keV gamma validation

4.1 Configuration and direct reference

The validation uses the 662 keV gamma line from global fuel 0 of cask 4, the assembly nearest to the nominal detector. The detector is the full nominal CLYC assembly: crystal, casing, photomultiplier, 50 mm lead collimator, 50 mm polyethylene collimator, and the polyethylene plug lip. The LiF collimator is disabled. The detector is at $(x, y, z) = (2930\,\text{mm}, 1580\,\text{mm}, 150\,\text{mm})$ with rotation $(0, -90, 0)^\circ$ in the nominal macro.

The direct ZLN file for cask 4 and global fuel 0 contains 5016 detector hits and a total deposited energy of 1048.78 MeV for a 180 s exposure. Scaling linearly to the 36 s exposure used here gives an expectation of 1003.2 hits and 209.756 MeV. The comparison is a closure test of the factorised workflow against a direct full-geometry calculation, not a new measurement.

The full gamma source contains 350 820 506 surface crossings in 23 GB, so loading it whole would exceed the workstation memory. The validation therefore uses a random one-million-event subset drawn from the 10^8-event source. The subset contains 3 484 873 crossings with total weight 0.012 982 2. With 1 000 000 source events and $T_{\text{gen}} = 36$ s the p1.5 normalisation gives

$$N_{p1.5} = (36\ \text{s})(1.4 \times 10^{14}\ \text{s}^{-1})\frac{0.0129822}{10^6} = 6.543 \times 10^7 \tag{7}$$

source events. The run uses eight threads and no biasing. The output file is written only after the event loop finishes; a partial file is not a valid source. The 36 s exposure is written as two independent 18 s chunks, `p15_gamma662_cask4_1Mevt_18s_z8_a.root` and `_b.root`. A single 36 s file reached 2.7 GB and its ROOT key directory was unreadable; each 18 s chunk remains at about 1.4 GB. Each chunk carries its own `sectorMeta` row with $T_{\text{gen}} = 18$ s; the p2 replays are run per chunk and summed.

4.2 p1.5 output and cell population

The two 18 s p1.5 chunks together produced 65 430 160 generated events, 32 715 080 per chunk. Each `sectorMeta` row records `nSectors`=8, `nZLayers`=8, `nCells`=64, `zMin`=−2040 mm, `zMax`=2040 mm, and `dz`=510 mm. Each `sectorFlux` tree contains one entry for each first crossing of a gamma into a cell. The entries are not spread uniformly over the 64 cells: the source assembly illuminates the cells that face it, and the azimuthal side entries are concentrated in the sectors adjacent to that direction.

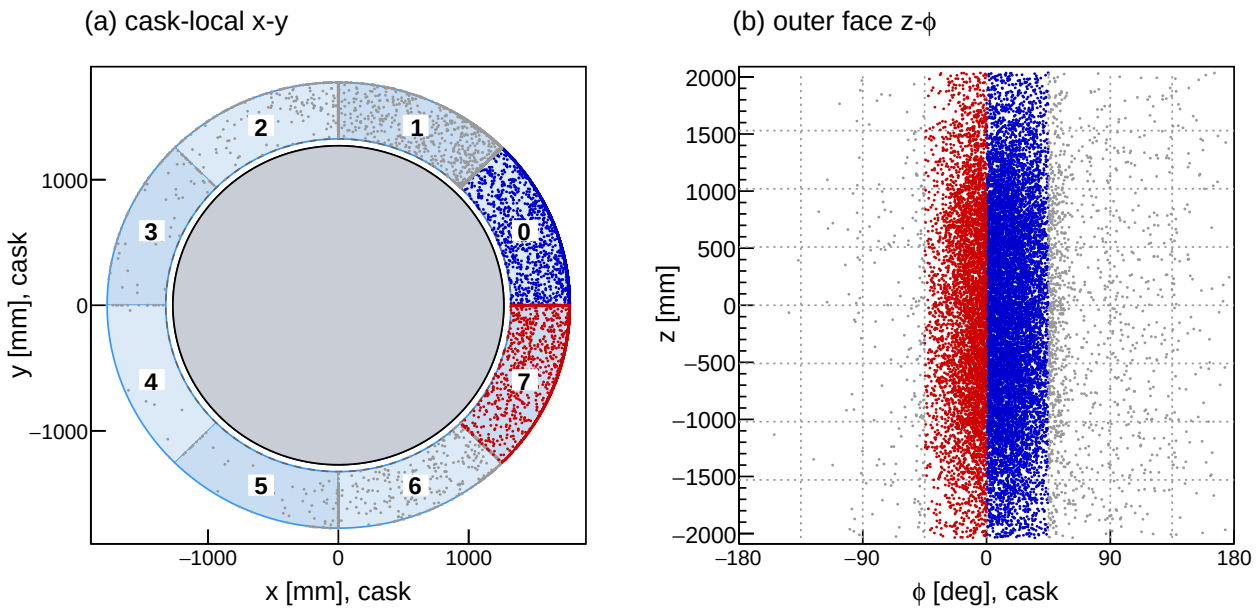


Figure 5: First-entry maps for the first 18 s chunk of the 64-cell p1.5 source. Left: cask-local x–y map, showing the eight sectors. Right: longitudinal position z versus azimuth ϕ for entries on the outer cylindrical face. The cell identifier is `zLayer` $\times\, 8 +$ `sector`. The source is one fuel assembly of cask 4, so the entries are localised in azimuth and in the longitudinal layer that faces the assembly. The blue and red entries mark sectors 0 and 7, the two sectors that contain the detector envelope. The left map uses a 10 000-entry subsample and the right map a 30 000-entry subsample for legibility; the histograms use all entries.

Figure 5 shows the cask-local x–y entry map and the outer-face z–ϕ map for the first 18 s chunk, and Figure 6 shows the cell population, face label, kinetic energy, longitudinal position, radial position, and polar angle. The second chunk has the same layout and is not plotted separately. The energy distribution is narrow because the source is a single gamma line plus the p1 smearing. The radial distribution fills the shell between the inner and outer radii, with the expected concentration near the inner surface. The longitudinal distribution shows that the full-height shell is populated at both ends; the entries near the cask top and bottom are real first crossings, not an artefact of a truncated scoring surface.

4.3 Replay result

The automatic cell selection returns cells 32 and 39. The replay loaded 3 936 600 entries for these cells and retained 3 806 673 unique source events after deduplication, so 129 927 entries were removed as repeated crossings of the same event. With $T_{\mathrm{replay}} = T_{\mathrm{gen}}$, Equation (6) gives 3 806 673 generated primaries. The replay produced 961 detector hits and a total deposited energy of 208.740 MeV.

A second replay used four cells, 24, 31, 32, and 39, to test whether the layer below the detector contributes a significant flux. It loaded 7 964 658 entries, retained 6 846 190 unique events, and generated 6 846 190 primaries. It produced 1016 hits and 212.392 MeV. The additional layer therefore changes the hit count by 5.7% and the deposited energy by 1.7%, which is consistent with the detector envelope being confined to layer 4 and with the small number of entries in the adjacent layer.

Table 2 compares the two replay results with the direct ZLN expectation. The two-cell replay gives a hit ratio of 0.958 and an energy ratio of 0.995. The four-cell replay gives 1.013 and 1.013. The two-cell sample contains about 953 independent hit events, so the statistical uncertainty on the hit count is approximately 3.2%. The two-cell and four-cell results bracket the direct yield; their difference is about 1.2 standard deviations and is therefore not significant at this sample size. They are not precise enough to exclude a few-percent bias, but they exclude a 10% normalisation error.

The 64-cell background model therefore reproduces the direct ZLN yield at the 3% statistical level for this configuration. The remaining difference is within the statistical uncertainty of this

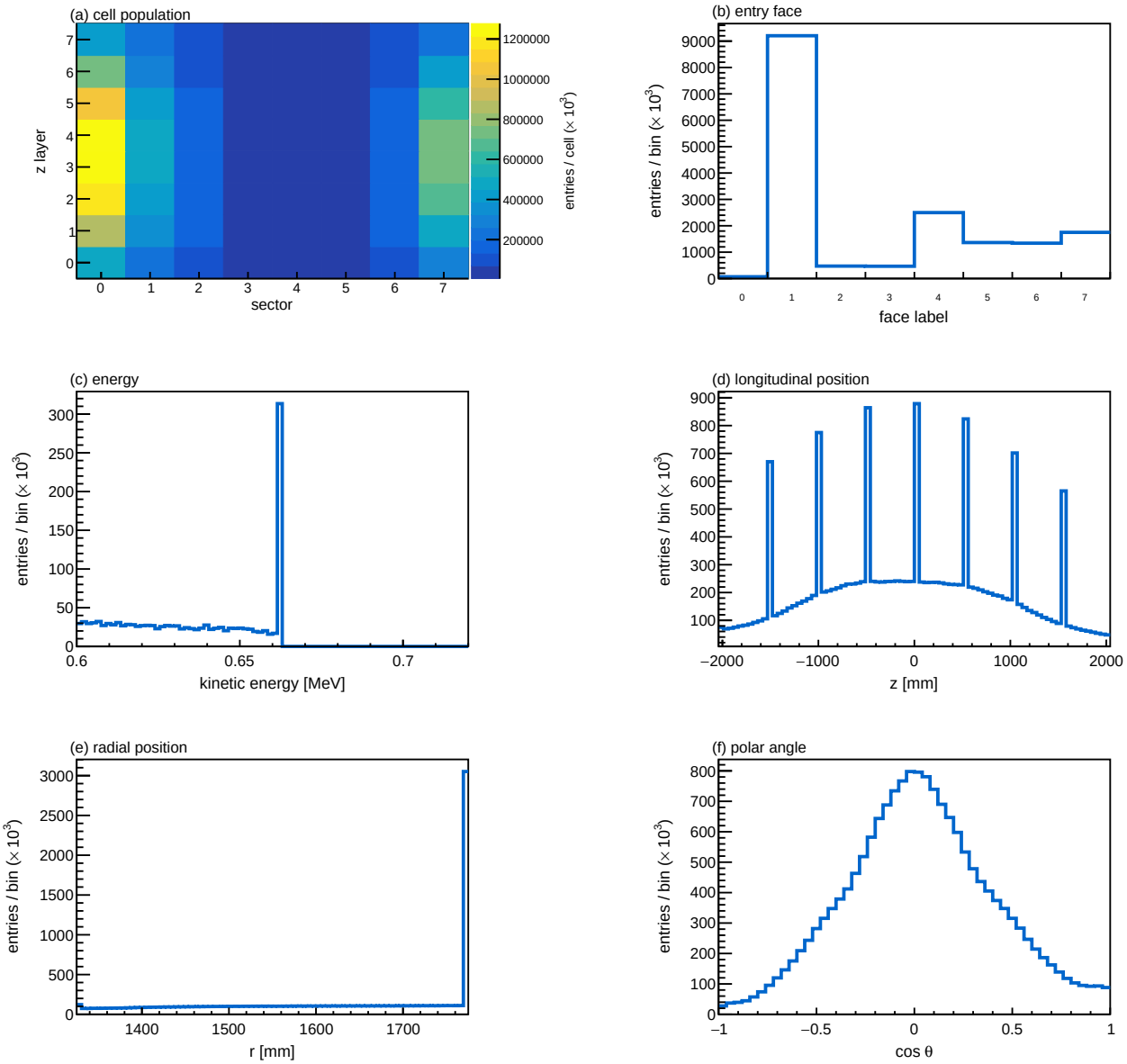


Figure 6: p1.5 `sectorFlux` distributions for the first 18 s chunk: cell population, entry face, kinetic energy, longitudinal position, radial position, and polar angle. Face labels 0–5 are the external inner, outer, top, bottom, and two azimuthal surfaces; labels 6 and 7 are the internal longitudinal boundaries between layers.

sample; a larger replay is needed to test for a few-percent bias.

Table 2: 662 keV gamma validation for cask 4 / global fuel 0. The direct ZLN yield is scaled from 180 s to the 36 s exposure. The p2 entry counts are the p1.5 entries after the cell filter and before and after event-level deduplication.

Replay	M_{loaded}	M_{unique}	N_{replay}	Hits	Deposited energy [MeV]
Direct ZLN, scaled to 36 s	—	—	—	1003.2	209.756
p2, cells 32 and 39	3 936 600	3 806 673	3 806 673	961	208.740
p2, cells 24, 31, 32, 39	7 964 658	6 846 190	6 846 190	1016	212.392

Figure 7 shows the corresponding energy-deposition spectra and the bin-by-bin ratio to the direct ZLN result.

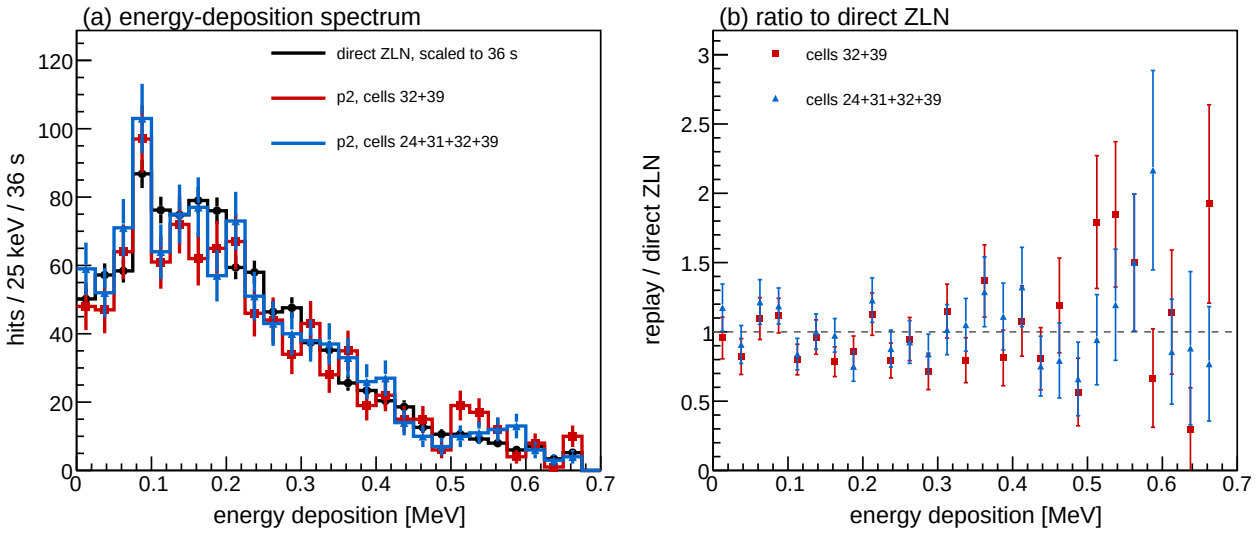


Figure 7: Energy-deposition spectra for the direct ZLN gamma-662 run and the 64-cell replay. The left panel shows absolute counts per 25 keV bin scaled to the 36 s exposure; the right panel shows the ratio of the replay to the direct ZLN spectrum. The replays are event-deduplicated. The two-cell selection uses the cells that contain the detector envelope; the four-cell selection adds the layer below it as a systematic check.

4.4 What the validation does and does not show

The validation shows that the 64-cell shell is populated, that the cell and face identifiers are written correctly, that the metadata row records the 64-cell layout, that the automatic cell selection returns the two cells that contain the nominal detector, and that the event-deduplicated replay produces a detector signal close to the direct ZLN result. It does not provide a background estimate. The source is one gamma line from one fuel assembly, the p1 file is a one-million-event subset, the detector is a single CLYC at one position, and the replay uses one candidate entry per source event. The 961-hit sample, with 953 independent hit events, has a 3.2% statistical uncertainty on the hit count; it is sufficient to exclude a 10% normalisation error but not to constrain the spectral shape at the few-percent level.

5 HPC production and validation plan

5.1 Production matrix

The target cask is cask 0. The other eleven casks are background sources, and each source cask is combined with six hexants. This gives

$$11 \text{ casks} \times 6 \text{ hexants} = 66 \tag{8}$$

independent p1.5 production configurations. The target cask remains passive in all of them. Table 3 groups the runs by cluster.

Table 3: p1.5 production matrix. Each row is multiplied by six hexants.

Source group	Background casks	p1.5 runs
Central cluster	3	18
$+x$ cluster	4	24
$-x$ cluster	4	24
Total	11	66

The runs are embarrassingly parallel: each configuration reads the same p1 file, builds the same cask array, and writes its own `sectorFlux` and `sectorMeta` output. A job array with one element

per `(cask, hexant)` pair is the natural mapping. The production macro must remove the one-million-event subset, use the full p1 source, and use the production T_{gen}. The target cask must be present in the geometry even when it is not the source, because the scoring shell is centred on it.

5.2 Cost model and HPC requirement

Let C_{cfg} be the measured wall-clock cost of one p1.5 configuration and $N_{\mathrm{cfg}} = 66$. The production cost is

$$C_{\mathrm{prod}} = 66\, C_{\mathrm{cfg}} \tag{9}$$

before accounting for queueing and failed-job resubmission. The replay stage adds

$$C_{\mathrm{replay}} = N_{\mathrm{scan}}\, C_{\mathrm{scan}}, \tag{10}$$

where a scan is one `(cell selection, detector, shield)` combination. The replay is cheaper per event than p1.5 because it starts from a unit-weight phase-space entry and does not repeat the cask-to-cell transport, but it is not free: the number of replayed primaries from Equation (6) can be large.

The measured 64-cell p1.5 rate on this eight-core workstation is about 15 000 evt/s for the 662 keV gamma source, so one 36 s configuration with 6.543×10^7 events takes about 73 min. The corresponding replay of 3.8×10^6 primaries takes a few minutes. These rates scale with the source type and the number of secondary particles; the neutron case is slower per event. The production campaign is therefore an HPC-scale workload, and the 66 independent p1.5 jobs map naturally to a job array. A workstation can produce a single validation configuration but should not serialise the full matrix.

5.3 Neutron validation plan

For neutrons, the closure test is more expensive but remains the production-relevant check. The same cask 4 / global fuel 0 configuration is the natural choice: cask 4 is at $(x, y) = (6240\,\mathrm{mm}, 1580\,\mathrm{mm})$, and its fuel-0 centre is at $(-662, +127)$ mm in the cask frame, i.e. on the detector-facing side. The direct ZLN file contains 1063 detector hits for 180 s, so it provides the requested $\mathcal{O}(10^3)$ count level.

The neutron p1.5 run has not yet been repeated with the 64-cell shell. The production validation must use the same cell selection and event deduplication as the gamma case, and it must record the number of entries on the two azimuthal side faces and on the internal longitudinal boundaries. Folding side entries into the radial faces would bias the face-resolved comparison. The replay must sample the cells that contain the detector; for the nominal position these are the sector-0 and sector-7 cells of layer 4.

5.4 Merging and metadata

The merger must preserve the per-cell entry count. A production file may contain one metadata row per cell, or the merger may add an explicit `M_cell` count alongside the existing total M. The replay sampler filters on `cellId` and computes M_{loaded} and M_{unique} from the filtered entries, so the physics normalisation is correct even if the metadata stores only the total; the per-cell count is still needed for bookkeeping and for detecting missing or duplicated files.

The merger should also check that the 64 cell identifiers are present or accounted for, that the shell dimensions and the number of longitudinal layers agree across files, that the target cask is the same in every file, and that no file was produced with a different p1 source or smearing configuration. The source cask, source type, hexant, global fuel, and batch identifier are already stored per entry and in the metadata.

5.5 Validation checklist

The following checks should be completed before the production result is used:

1. **Geometry containment.** Verify that the detector and shield lie between the inner and outer faces, do not overlap the cask body, and that the hemi-shield variant remains inside the shell and clear of the inner face. For the nominal detector at $\varphi = 0°$, note that it lies on the boundary between sectors 0 and 7; both cells must be selected.

2. **Cell coverage.** Verify that every populated cell has a non-zero entry count, that the cell identifiers are unique, and that the sum of the per-cell counts equals the total entry count. Check the two external caps and the internal longitudinal labels separately.

3. **First-entry correctness.** Compare the number of entries with and without the boundary-status check and with a step-level reference count; confirm that the 256-bit mask suppresses duplicates without killing tracks.

4. **Deduplication.** Compare the event-deduplicated replay with a track-aware replay for the same source sample. The difference measures the bias introduced by keeping only the earliest entry per event. At minimum, report M_{loaded}, M_{unique}, and the fraction removed for every cell selection.

5. **Phase-space closure.** Repeat the p1.5-to-p2 closure test for every populated cell and for both particle types. Check energy, polar angle, azimuth, and all face categories.

6. **Symmetry.** Rotate a source configuration by 60° (hexant) or 45° (sector) and compare the cell populations after the corresponding rotation. Deviations should be consistent with the geometry, not with a bookkeeping error.

7. **Normalisation.** Verify Equation (6) against a direct p2 run for the same source and time. The ratio of hit yields should be consistent with one within statistics. Report the deduplication prescription and the selected cells with every ratio.

8. **Reproducibility.** Record the executable commit, the p1 file checksum, the macro, the random seed, and the production T_{gen} for every output file.

6 Limitations and next steps

The main limitations of the pilot are the single gamma line, the single source cask, the one-million-event source subset, the single detector position, and the event-level deduplication. The production campaign must address all of these. In addition, the current metadata stores the total M; the merger should add the per-cell count before the production files are archived. The gamma validation used a one-million-event random subset because the full source exceeds the workstation memory; the subset normalisation is correct, but the full source should be used in production.

The deduplication is the most important modelling approximation. It keeps the earliest entry per source event, which removes repeated crossings of the artificial cell boundaries but also discards every later entry of the same event. If a later entry is closer to the detector, the replay misses it. The present gamma test is consistent with the direct ZLN result at the approximately 3% statistical level, but it cannot distinguish the two-cell and four-cell selections, whose hit yields differ by 5.7%. A track-aware replay, or a replay that carries all cell entries with an explicit per-entry weight, is required before the 64-cell phase space is used for a production background estimate.

The geometry near the inner radius also deserves a dedicated check. The sector inner radius is close to the fin tips, and a detector or shield placed at the nominal radius could, in some configurations, approach the inner face. The shell is a parallel-world scoring surface, so an intersection would not change the transport, but it would change the interpretation of the first-entry sample. The containment check should be part of the production qualification.

Finally, the OpenGL renders in this report are illustrative. They omit the hall and hide internal cask details. The dimensioned plan view and the radial–longitudinal cross-section remain the reference figures for geometry.

7 Conclusion

The 64-cell background-sector workflow is implemented and exercised end to end. The p1.5 stage scores first entries into a parallel-world shell of eight azimuthal sectors and eight longitudinal layers, the p2_bkgd stage replays the unit-weight phase space through the detector geometry, and the 662 keV gamma check reproduces the direct ZLN yield at the approximately 3% statistical level. The event-level deduplication removes repeated crossings of the artificial cell boundaries, but it keeps only one candidate entry per source event; a track-aware replay is therefore required before the nominal detector position is validated at the few-percent level. The production campaign requires 66 independent p1.5 configurations and a set of replay scans; this is an HPC-scale workload, but it is embarrassingly parallel and can be mapped to a job array. The next step is to remove the one-million-event subset, add per-cell metadata, run the production matrix, and complete the validation checklist.

A Reproducing the validation

The p1.5 validation requires the background-sector mode on the command line because the parallel world must be registered before the macro runs:

```
./build/G4DCSmonitor --mode background-sector \
  macros/validation/p15_gamma662_cask4_1Mevt_18s_a.mac
./build/G4DCSmonitor --mode background-sector \
  macros/validation/p15_gamma662_cask4_1Mevt_18s_b.mac
```

The p2 replay uses the same executable without `--mode`:

```
./build/G4DCSmonitor \
  macros/validation/p2_gamma662_cask4_1Mevt_18s_a.mac
./build/G4DCSmonitor \
  macros/validation/p2_gamma662_cask4_1Mevt_18s_b.mac
./build/G4DCSmonitor \
  macros/validation/p2_gamma662_cask4_1Mevt_18s_4cell_a.mac
./build/G4DCSmonitor \
  macros/validation/p2_gamma662_cask4_1Mevt_18s_4cell_b.mac
```

The two 18 s p2 outputs are combined with `hadd`:

```
hadd -f build/validation/p2_gamma662_cask4_1Mevt_36s_z8.root \
  build/validation/p2_gamma662_cask4_1Mevt_18s_z8_a.root \
  build/validation/p2_gamma662_cask4_1Mevt_18s_z8_b.root
hadd -f build/validation/p2_gamma662_cask4_1Mevt_36s_4cell_z8.root \
  build/validation/p2_gamma662_cask4_1Mevt_18s_z8_4cell_a.root \
  build/validation/p2_gamma662_cask4_1Mevt_18s_z8_4cell_b.root
```

The validation macros use cask 4 and global fuel 0, set the 64-cell layout with the command

```
/dcs-monitor/det/setSectorZLayers 8
```

and write files with the `_z8` suffix. The source file is `surface_gamma662_cask4_1Mevt.root` in `build/validation`. The figures are generated from the repository root with

```
root -l -b -q report/sector_implementation/scripts/make_geometry_figs.C
root -l -b -q report/sector_implementation/scripts/make_data_figs.C
./report/sector_implementation/scripts/make_ogl_figs.sh
```

The data figures use the first 18 s p1.5 chunk by default; set `P15_FILE` to another chunk to reproduce the maps from that file. The ROOT scripts crop the PDF page boxes with `pdfcrop`; the OGL driver exports vector PDFs and rasterises them for the report.